



Interior curves

From four boundary chains, make interior U and V curves that are guaranteed to cross -- a ready network for Gordon Surface.



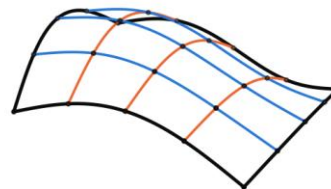
What it does

A Gordon network needs curves that actually cross. Interior curves fill a four-sided boundary with the rows and columns of a Coons patch, which share their grid points by construction.

Use it when

- Start a Gordon network from its boundary
- Need crossing interior curves to edit later

Interior curves: 4 boundaries (black) -> Coons grid -> 3 interior U (blue) and 3 V (orange), sharing the grid points



4 boundaries -> Coons grid -> 3 + 3 interior curves.



The dialog

IMAGE MISSING:
docs/decks/interior_curves/shots/dialog.png

#	Parameter	What it means
1	U0	Boundary chain (edges joined).
2	U1	Opposite boundary.
3	V0	Crossing boundary.
4	V1	Opposite crossing boundary.
5	Interior U cure count	Interior U curves (3; 1-15).
6	Interior V cure count	Interior V curves (3).
7	Debug & Details	Print curve data (its output tolerance / degree are not used).



Outputs and messages

Outputs

Key	What it holds
interiorU_i, interiorV_i	The interior wires.

Messages

	Message / what to do
Error	Boundary curves don't form a closed loop at v0



Tips and limits



Coarse

Each curve is fitted through count + 2 points (3 x 3 -> 5 points each).



Uniform in parameter

Not in length; prototype, untested.



Where it lives

Document	<code>gordonSurface (d0950ed72a894fbe35c27d43)</code>
Tab	<code>interiorCurves (symbol myFeature)</code>
Tests	none yet
Examples	Test Part Studio 1: Interior curves 1
Explainer	<code>docs/explainers/gordon_surface/gordon_surface_explained.md</code> , section 2.2
Related	Gordon Surface, Interior curves, Scaled Curve, Modify curve end, Pull surface, Simplify surface